Hospital Emergency Room Data Analysis Report

Data Analysis Project

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Executive Summary

This report presents a comprehensive analysis of hospital emergency room (ER) data. The objective of this project is to evaluate patient demographics, operational efficiency, and quality of service indicators while generating strategic insights to support data-driven decision-making.

Introduction

Background

Emergency departments play a critical role in healthcare systems. Understanding patient trends and operational efficiency is essential for delivering high-quality care.

Project Objectives

- Analyze patient demographics
- Evaluate operational efficiency
- Assess quality of service
- Generate strategic insights for decision-making

Methodology

Data Source

The dataset includes hospital emergency room records covering patient demographics, wait times, departmental referrals, flagged admissions, and satisfaction scores.

Analytical Tools

Data analysis was conducted using Python, SQL, Power BI, and statistical methods including averages, correlations, and trend analysis.

Data Cleaning

Data preprocessing included handling missing values, validating records, and addressing NULL satisfaction scores.

Dataset Summary

Dataset size:

- **Number of Rows:** 9216
- **Columns:** 12

Key Features:

- **Patient Demographics:** (Age, Gender, Full Name, Race, Patient ID)
- **Admission & Referral details:** (Admission Date, Department Referral, Admission Flag, Clinical Management Status)
- **Operational & Feedback Metrics:** (Wait Time, Wait Time Z-Score, Satisfaction Score)
- **Missing Data:** 6,699 missing values in the patient_satisfaction_score column

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9216 entries, 0 to 9215
Data columns (total 12 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   patient_id                    9216 non-null   int64
 1   patient_fullname              9216 non-null   object
 2   patient_admission_date        9216 non-null   datetime64[ns]
 3   patient_gender                9216 non-null   object
 4   patient_age                   9216 non-null   int64
 5   patient_race                  9216 non-null   object
 6   department_referral           9216 non-null   object
 7   patient_admission_flag        9216 non-null   bool
 8   patient_satisfaction_score    2517 non-null   float64
 9   patient_waittime              9216 non-null   int64
 10  patients_clinical_management  9216 non-null   bool
 11  waitime_zscore                9216 non-null   float64
dtypes: bool(2), datetime64[ns](1), float64(2), int64(3), object(4)
memory usage: 738.1+ KB
```

Statistical Analysis

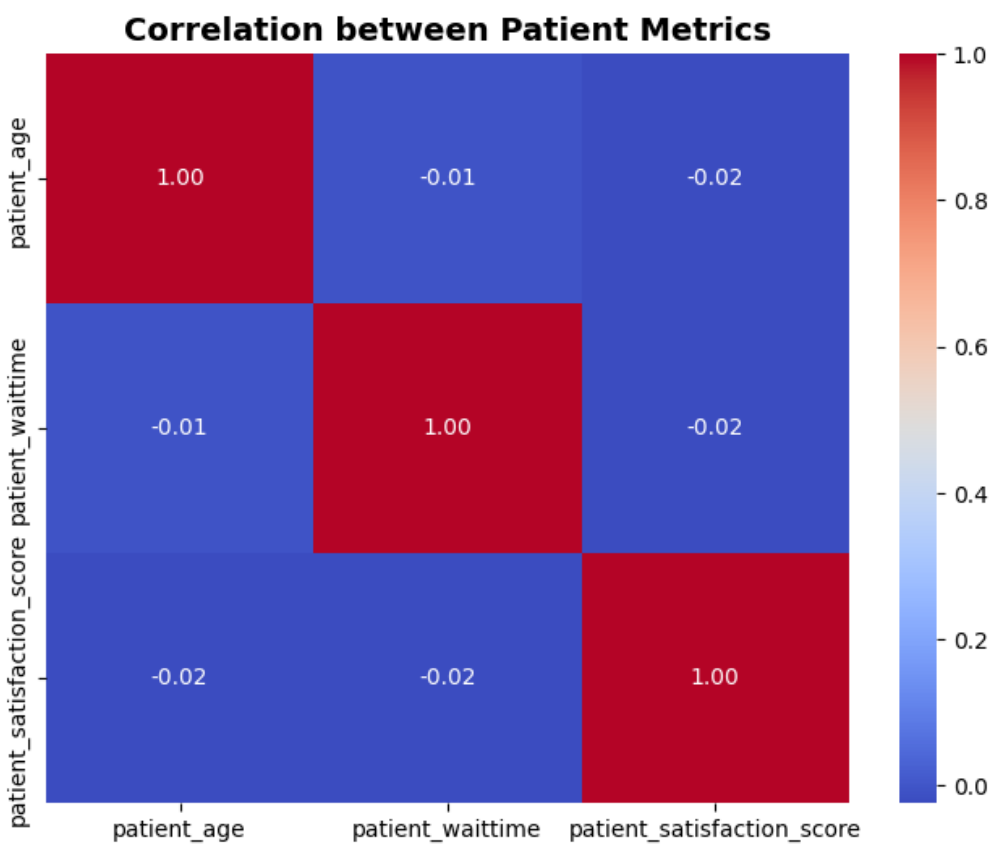
	patient_id	patient_admission_date	patient_age	patient_satisfaction_score	patient_waittime	waitime_zscore
count	9.216000e+03	9216	9216.000000	2517.000000	9216.000000	9.216000e+03
mean	5.022227e+08	2024-01-16 14:21:23.372395776	39.855143	4.992054	35.259874	1.079383e-16
min	1.000440e+08	2023-04-01 01:13:00	1.000000	0.000000	10.000000	-1.714240e+00
25%	3.011016e+08	2023-08-24 14:19:30	20.000000	2.000000	23.000000	-8.320058e-01
50%	5.022989e+08	2024-01-17 06:50:30	39.000000	5.000000	35.000000	-1.763613e-02
75%	7.039526e+08	2024-06-09 11:55:15	60.000000	8.000000	48.000000	8.645977e-01
max	8.999825e+08	2024-10-30 23:44:00	79.000000	10.000000	60.000000	1.678967e+00
std	2.308648e+08	NaN	22.755125	3.138043	14.735323	1.000000e+00

1. Anomaly Detection:

(Identification of outliers as values deviating by more than 3 standard deviations)

Missing Data: 0 outliers identified (indicating all wait time data points fall within a statistically normal range).

2. Correlation Matrix

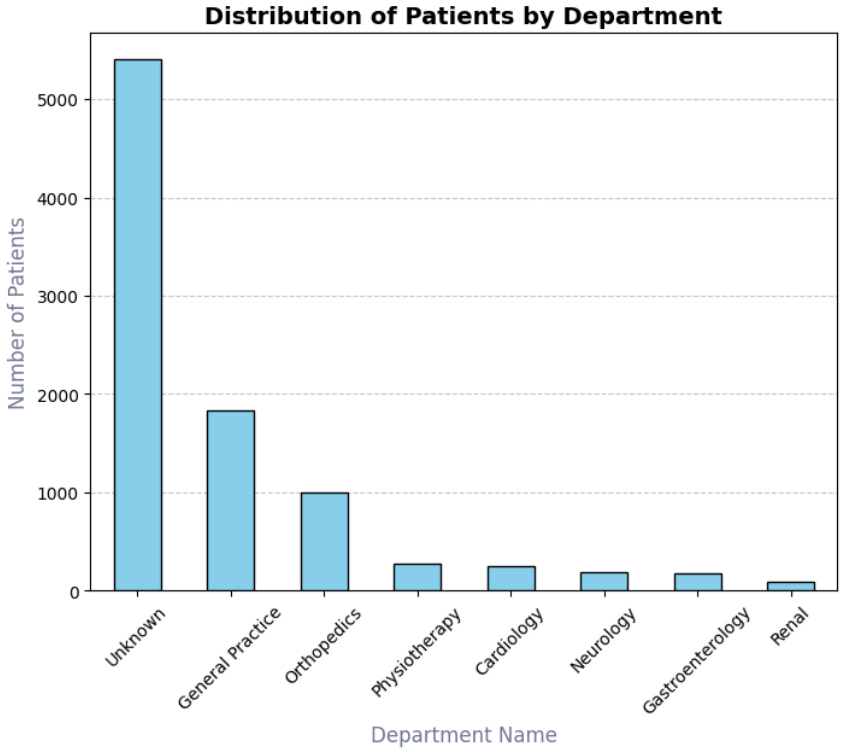


Zero Correlation: There is effectively **no relationship** between patient age, wait time, and satisfaction scores, as all correlation coefficients are nearly zero (-0.01 to -0.02).

Independent Variables: Changes in wait time do not appear to influence satisfaction scores in this dataset, suggesting other factors are driving patient sentiment.

Statistical Uniformity: The data is highly uncorrelated, indicating that these specific metrics act independently of one another across the entire patient population.

3. Advanced Distribution

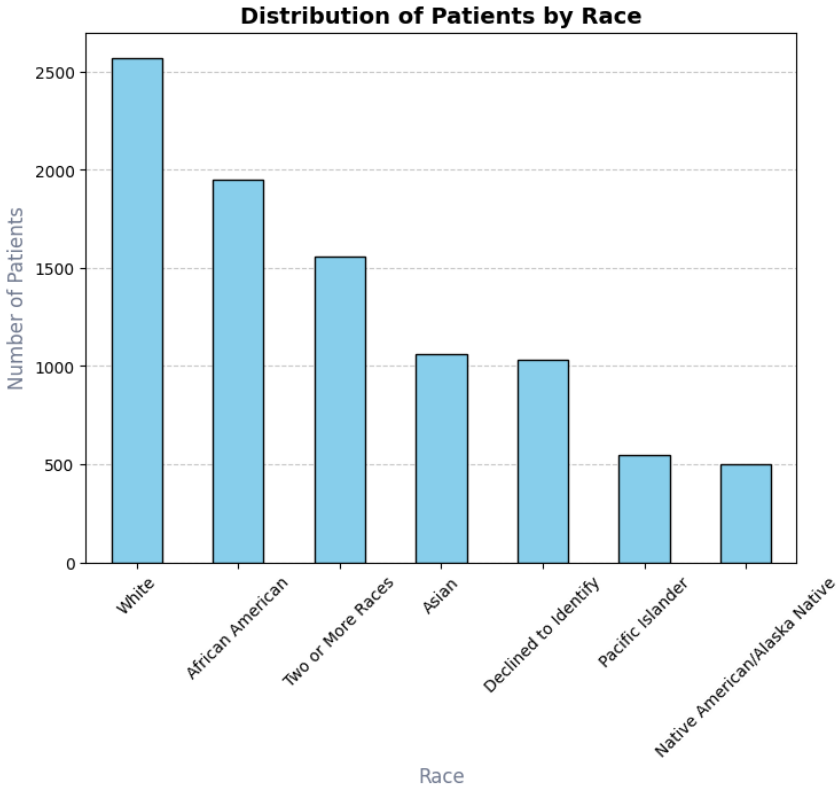


Department Categories: (Unknown, General Practice, Orthopedics, Physiotherapy, Cardiology, Neurology, Gastroenterology, Renal)

Patient Volume Details: (Number of Patients per Department)

Distribution Behavior: (High concentration in Unknown/GP, low volume in specialized units like Renal)

Missing Data: Over **5,000 patients** categorized as "Unknown," representing more than half of the total dataset.

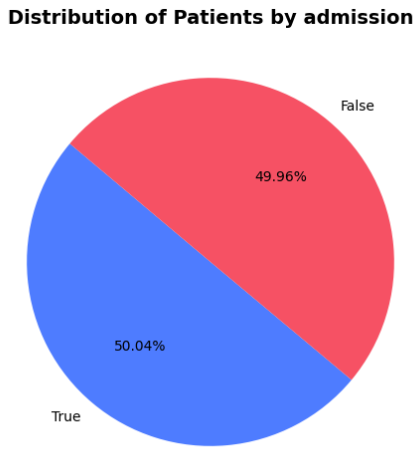


Customer demographics: (White, African American, Two or More Races, Asian, Pacific Islander, Native American/Alaska Native)

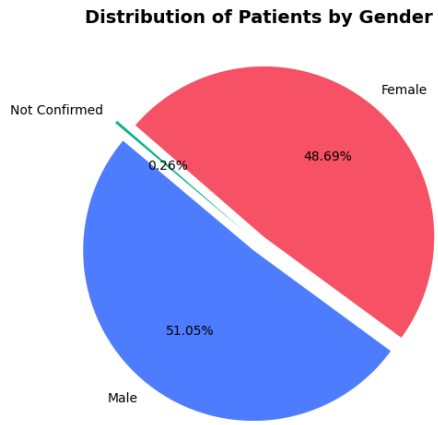
Patient Volume details: (Number of Patients per Race category ranging from ~500 to ~2500)

Patient Behavior: (Significant group of approximately 1,000 patients "Declined to Identify" their race)

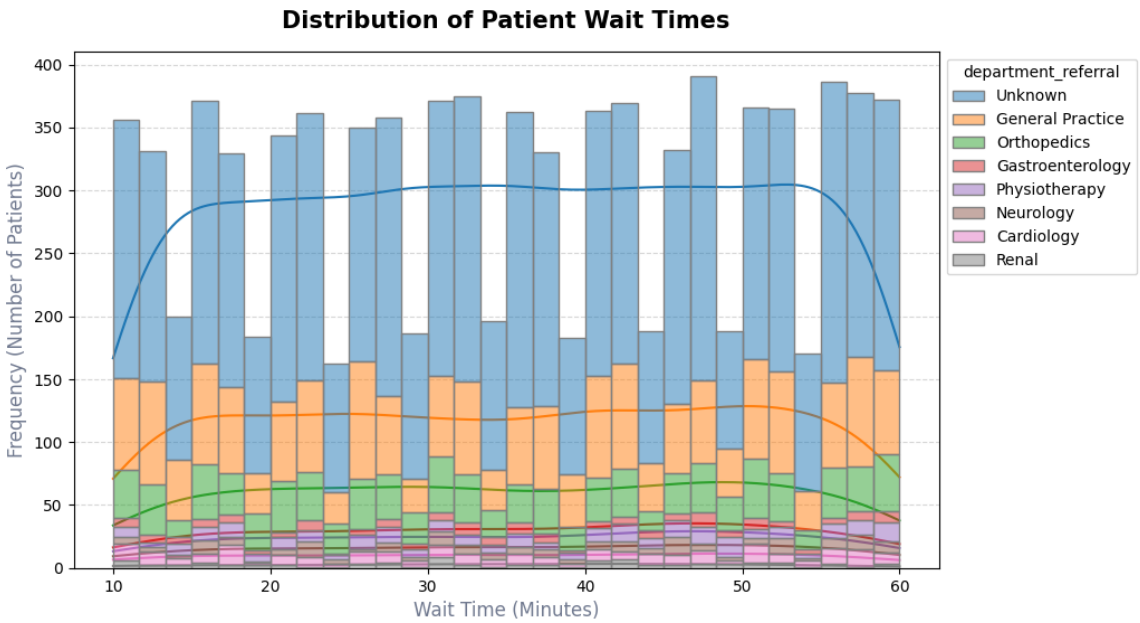
Missing Data: 11% of the total dataset is categorized as "Declined to Identify," creating a gap in precise demographic mapping.



Distribution Behavior: (Nearly perfect 50/50 split between flagged and non-flagged admissions).



Distribution Behavior: (Balanced split with a slight majority of **Male** patients at **51.05%** versus **Female** at **48.69%**).

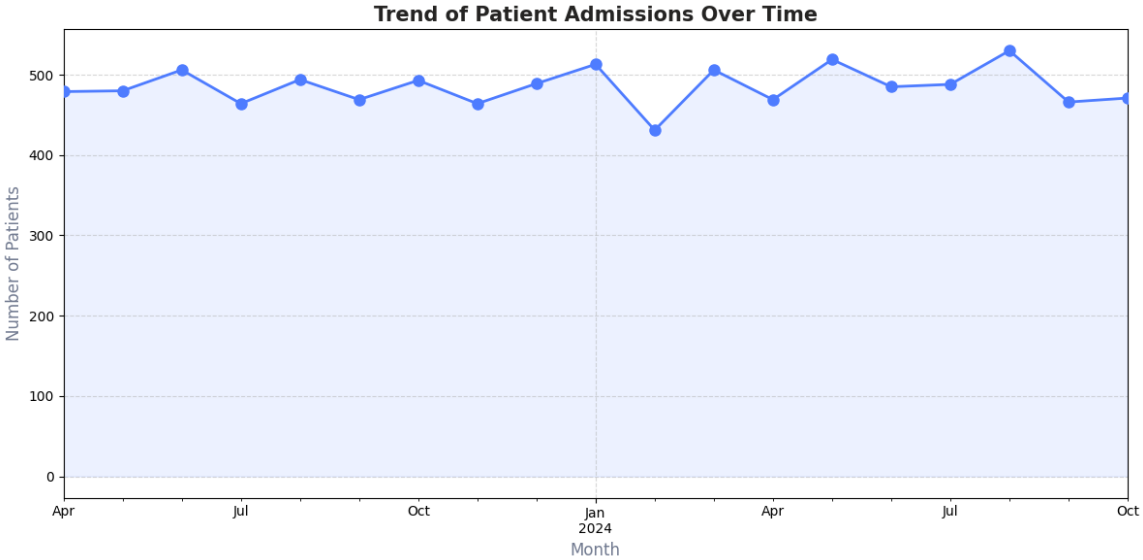


Operational Details: (Wait Time in Minutes, Number of Patients)

Department Referral details: (Unknown, General Practice, Orthopedics, Gastroenterology, Physiotherapy, Neurology, Cardiology, Renal)

Wait Time Behavior: (Uniform distribution across the 10-60 minute range, with consistent patient volume at nearly every 5-minute interval)

Missing Data: A massive portion of the wait time data is categorized under **"Unknown"** (the largest blue bars), obscuring department-specific efficiency trends.



Time Period Details: (Month, Year spanning from April 2023 to October 2024)

Admission Volume details: (Number of Patients per month, ranging from approximately 430 to 530)

Admission Behavior: (Relatively steady trend with cyclical fluctuations and a noticeable peak in August 2024)

Missing Data: 0 values (The time series provides a continuous record of admissions for the 19-month period shown)

Patient Demographics & Trends

Age Distribution

Analysis of patient age groups to identify high-demand segments.

	age_category	Number_of_patients	percentage
1	Adults	4799	52%
2	Seniors	2307	25%
3	Children	2110	22%

Adults are the primary high-demand segment, accounting for more than half of the total patient base at **52%**.

Seniors and Children make up the remaining portion, representing **25%** and **22%** of patients, respectively.

The total patient volume is **9,216**, with a clear concentration of demand within the adult demographic.

Department Referrals

Evaluation of departments receiving the highest ER referrals.

	department_referral	number_of_referals	percentage
1	Unknown	5400	58%
2	General Practice	1840	19%
3	Orthopedics	995	10%
4	Physiotherapy	276	2%
5	Cardiology	248	2%
6	Neurology	193	2%
7	Gastroenterology	178	1%
8	Renal	86	0%

Data Gaps: A significant majority (**58%**) of ER referrals are categorized as **"Unknown,"** indicating a major need for better data collection or documentation.

Top Known Departments: General Practice and **Orthopedics** are the primary identified recipients, accounting for **19%** and **10%** of referrals, respectively.

Specialized Care: Other specialized units like Cardiology and Neurology receive minimal referrals, each making up **2% or less** of the total volume

Diversity Profile

Assessment of gender and demographic distribution of patients.

	patient_race	number_of_patients	percentage
1	White	2571	27%
2	African American	1951	21%
3	Two or More Races	1557	16%
4	Asian	1060	11%
5	Declined to Identify	1030	11%
6	Pacific Islander	549	5%
7	Native American/Alaska Native	498	5%

Primary Demographics: The largest patient groups are **White (27%)** and **African American (21%)**, together making up nearly half of the total population.

Diverse Representation: Individuals identifying as **Two or More Races** or **Asian** represent a significant portion, totaling **27%** of the patient base.

Data Specifics: While **11%** declined to identify their race, Pacific Islanders and Native Americans each account for the smallest share at **5%** each

Operational Efficiency Analysis

Average Wait Time by Department

Comparison of mean wait times across hospital departments.

	department_referral	average_waittime	total_patients
1	Neurology	36.8	193
2	Physiotherapy	36.57	276
3	Gastroenterology	35.83	178
4	Cardiology	35.35	248
5	Unknown	35.29	5400
6	Orthopedics	34.98	995
7	General Practice	34.91	1840
8	Renal	34.7	86

Wait Time Range: Average wait times are remarkably consistent across all departments, ranging from a high of **36.8 minutes** in Neurology to a low of **34.7 minutes** in Renal.

Volume Imbalance: The **"Unknown"** category contains the vast majority of patients (**5,400**), dwarfing identified departments like General Practice (1,840) and Orthopedics (995).

Efficiency Paradox: Despite having the fewest patients (86), the **Renal** department is only about two minutes faster than **Neurology**, which has over double the patient volume.

Wait Time vs Patient Age

Correlation analysis between patient age and waiting duration.

	age_category	total_patients	avg_wait_time	min_wait	max_wait
1	Adults	4799	35.32	10	60
2	Children	2110	35.32	10	60
3	Seniors	2307	35.07	10	60

Consistent Wait Times: Average wait times are nearly identical across all age groups, holding steady at approximately **35 minutes**.

Uniform Limits: Every category shares the exact same minimum (**10 min**) and maximum (**60 min**) wait durations, suggesting a standardized process regardless of age.

Volume Distribution: While wait times are uniform, **Adults** make up the bulk of the data with **4,799 patients**, more than doubling the volume of Children or Seniors.

Flagged Admissions Percentage

Calculation and interpretation of flagged admission rates.

	flagging_percentage
1	50.0434%

Flagging Rate: Roughly half of all admissions are being flagged, with a precise rate of **50.0434%**.

Operational Volume: This high percentage suggests that **one out of every two** patient admissions requires additional review or meets specific "flag" criteria.

Metric Significance: The near-perfect **50/50 split** indicates a potentially significant administrative workload dedicated to investigating or processing these flagged cases.

Quality of Service Analysis

Wait Time Impact on Satisfaction

Evaluation of how waiting duration affects satisfaction scores.

	feedback	avg_wait_time	total_responses
1	Detractor	35.52	1600
2	Promoter	35.26	443
3	Unknown	35.23	6699
4	Passive	34.87	474

Negligible Impact: There is almost no correlation between wait time and satisfaction, as all groups have nearly identical average wait times of approximately **35 minutes**.

Volume Distribution: The **"Unknown"** feedback category dominates with **6,699 responses**, suggesting that the vast majority of patients are not providing specific satisfaction ratings.

Sentiment Comparison: Detractors (1,600 responses) actually have the highest average wait time (**35.52 min**), while **Passives** (474 responses) have the lowest (**34.87 min**), though the difference is only about 39 seconds.

NULL Satisfaction Rate

Assessment of missing satisfaction score responses.

	total_responses	null_count	null_rate
1	9216	6699	72.00%

Significant Data Gap: There is a critical lack of feedback, with **72%** of all satisfaction score responses being missing or null.

Low Participation: Out of **9,216 total responses**, only **2,517** actually provided a score, while **6,699** remained empty.

Reporting Challenge: This high null rate suggests that any current satisfaction insights may be unreliable due to the vast majority of patients not providing feedback.

Satisfaction by Department

Comparison of patient satisfaction across departments.

	department_referral	average_satisfaction_score	number_of_serveys	missing_scores
1	Gastroenterology	5.8	54	124
2	Neurology	5.28	53	140
3	Cardiology	5.14	71	177
4	General Practice	5.06	503	1337
5	Physiotherapy	4.99	83	193
6	Unknown	4.95	1440	3960
7	Orthopedics	4.86	290	705
8	Renal	4.57	23	63

Top Performance: Gastroenterology leads with the highest average satisfaction score of **5.8**, while the **Renal** department has the lowest at **4.57**.

Volume & Participation: The **"Unknown"** category dominates with **1,440 surveys**, but also suffers from a massive **3,960 missing scores**, highlighting a major data collection issue.

Identification Gap: General Practice has the highest survey count among known departments (**503**), yet more than **72%** of its potential satisfaction scores are missing.

Strategic Business Insights

Patient Loyalty & Frequency

Analysis of repeat visits and patient frequency trends.

	patient_fullname	frequency
1	A Gadsdon	2
2	B Champneys	2
3	F Dienes	2
4	F Fowlston	2
5	F Keogh	2
6	H Kehoe	2
7	J Coey	2
8	J Flarity	2
9	J Franceschino	2
10	J Piggen	2
11	J Rollinson	2
12	J Schermick	2
13	K McKeon	2
14	L Lovick	2
15	L Pacht	2
16	L Torald	2
17	O Winny	2
18	P Lotte	2
19	Q Radish	2
20	Q Simon	2
21	R Eede	2
22	R Mallia	2
23	T Le Clercq	2
24	T Smithies	2
25	U Bulfield	2
26	U Pharaoh	2
27	V Kinzel	2
28	W Keyworth	2
29	X Lewsy	2
30	Z Gritten	2
31	Z Row	2

Uniform Frequency: Every patient listed in this segment shows a consistent frequency of **2 visits**, indicating a base of repeat "loyal" patients.

Diverse Patient Base: The list contains **31 unique patients** from various alphabetical groupings, showing that loyalty is spread across a wide range of individuals.

Engagement Indicator: While these are repeat visitors, the data is capped at 2, suggesting these may be **short-term follow-ups** rather than long-term chronic care patients.

Wait Time Consistency

Examination of variability in wait times across departments.

	department_referral	wait_time_std_dev
1	Renal	14.42
2	Physiotherapy	14.62
3	Unknown	14.62
4	Gastroenterology	14.66
5	Neurology	14.66
6	Orthopedics	14.82
7	General Practice	14.98
8	Cardiology	15.47

High Consistency: The standard deviation of wait times is remarkably similar across all departments, ranging narrowly from **14.42** to **15.47**.

Most Stable: The **Renal** department shows the least variability (**14.42**), suggesting its wait times are the most predictable for patients.

Most Variable: Cardiology has the highest standard deviation (**15.47**), indicating it experiences the most fluctuation in wait times compared to other units.

Time Trend Analysis

Identification of seasonal patterns and long-term ER trends.

	admission_month	total_admission	avg_wait_time	avg_satisfaction_score	mom_change
1	2023-04	479	34.9	5.3	NULL
2	2023-05	480	34.43	5.16	1
3	2023-06	506	35.58	5.18	26
4	2023-07	464	34.72	4.99	-42
5	2023-08	494	36.39	5.06	30
6	2023-09	469	34.27	4.98	-25
7	2023-10	493	34.93	4.74	24
8	2023-11	464	35.19	5.09	-29
9	2023-12	489	34.76	4.68	25
10	2024-01	513	36.32	4.96	24
11	2024-02	431	36.67	4.72	-82
12	2024-03	506	35.88	5.33	75
13	2024-04	469	35.04	4.63	-37
14	2024-05	519	35.81	5.15	50
15	2024-06	485	35.51	4.71	-34
16	2024-07	488	35.2	4.79	3
17	2024-08	530	35.11	5.18	42
18	2024-09	466	35.06	4.91	-64
19	2024-10	471	34.05	5.31	5

Steady Volume with Spikes: Total admissions consistently fluctuate between **431 and 530 per month**, with notable peaks in August 2024 (530) and May 2024 (519).

Wait Time Stability: Average wait times remain highly stable, hovering narrowly between **34 and 36 minutes** regardless of the admission volume.

Satisfaction Fluctuations: Average satisfaction scores vary between **4.63 and 5.33**, with the highest satisfaction recorded in March 2024 and October 2024.

Key Findings

• Identification of high-demand age groups

• Departments experiencing longer wait times

• Relationship between wait time and satisfaction

• Data completeness considerations

Recommendations

1. Data Integrity & Governance

- **Recommendation:** Implement a **Standardized Intake Protocol** to eliminate "Unknown" categories, which currently represent 58% of referrals and over half of all departmental data.
- **Impact:** Ensures resource allocation is based on accurate departmental needs rather than skewed or missing information.
- **Action:** Transition from free-text entry to mandatory dropdown menus in the Electronic Health Record (EHR) for all referral and demographic fields.

2. Patient Feedback & Engagement

- **Recommendation:** Deploy a **Multi-Channel Satisfaction Strategy** to address the 72% null rate in satisfaction scores.
- **Impact:** Provides a representative sample of patient sentiment, moving beyond the current "Unknown" feedback dominance (6,699 missing scores).
- **Action:** Integrate automated SMS-based surveys triggered immediately upon discharge and offer small incentives for survey completion to increase the 27% response rate.

3. Operational Scaling & Staffing

- **Recommendation:** Utilize **Predictive Demand Modeling** based on the "Trend of Patient Admissions Over Time" chart to manage monthly fluctuations (430–530 patients).
- **Impact:** Maintains the consistent 35-minute wait time even during peak months like August and May.
- **Action:** Align staff scheduling and "On-Call" rotations with identified peak cycles to prevent burnout and maintain operational efficiency.

4. Target Demographic Alignment

- **Recommendation:** Develop a **Geriatric and Adult-Centric Care Model**, as these groups constitute 77% of the total patient base.
- **Impact:** Tailors the hospital's environment and clinical expertise to the primary users of the facility.
- **Action:** Invest in specialized training for staff in age-related chronic condition management and ensure facility accessibility meets the needs of the 25% senior population.

5. Quality Control & Outlier Management

- **Recommendation:** Establish an **Automated Flagging Review Board** to analyze the 50.04% of flagged admissions.
- **Impact:** Reduces the administrative burden of review if the "Flagged" criteria are currently set too broadly.
- **Action:** Audit the "Admission Flag" logic to ensure it targets high-risk cases effectively rather than triggering for 1 out of every 2 patients.

6. Data & Performance

- Establish performance dashboards

Conclusion

This report provides a structured evaluation of emergency room performance. The findings support strategic improvements in operational efficiency and service quality.